

Project Proposal

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Title of the Project: Machine Learning based Land Use and Land Cover Classification and Zoning Using Sentinel Imagery of Hyderabad Region.

Problem Statement and Practical Relevance:

Land Use and Land Cover (LULC) information with reliable, up-to-date and accurate analysis is crucial for informed decision-making in various sectors such as agriculture, urban planning, water resource management, disaster management, monitoring of land encroachments and land degradation, and overall environmental health. India being a densely populated country, understanding LULC patterns is essential for efficient use of land resources.

In rapidly developing areas such Hyderabad and surrounding regions in Telangana, land use is undergoing numerous changes due to urban expansion, agricultural modernization, and infrastructural development. Traditional field surveys, which are time consuming & resource intensive, along with obsolete maps, are insufficient to meet the growing demand for real-time, precise, and scalable solutions. While satellite-based classification provides a reliable approach, its effective use requires advanced tools and techniques for data preprocessing, feature extraction, classification, and spatial analysis. This research project aims to develop a Machine Learning integrated remote sensing framework by leveraging advanced geospatial technologies such as Google Earth Engine, SNAP and GIS tools. The applications are known to provide powerful capabilities for image correction, feature extraction, classification, and change analysis for satellite imagery.

Objectives of the Study:

1. To collect and preprocess Sentinel-2 satellite imagery.
2. To perform LULC classification using machine learning and image-analysis.
3. To prepare zoning maps (agriculture, built-up, vegetation, forest, water bodies, barren land, etc.) derived from the classification results.
4. To evaluate classification accuracy using confusion matrices and validation samples.
5. To analyse spatial land use changes using multi-temporal satellite imagery.
6. To assess vegetation health through NDVI and other relevant spectral indices.
7. To generate geospatial outputs suitable for integration into GIS platforms for planning and policymaking.

Literature review:

Satellite-based remote sensing has become a standard technique for land classification due to its wide coverage and high temporal resolution.[1] Machine learning methods integrated with geospatial tools, such as Maximum Likelihood, Random Forest, and SVM, have proven effective in distinguishing built-up regions, vegetation, water bodies, and barren lands.[2][3] The SNAP toolbox is widely used for atmospheric correction (Sen2Cor), speckle filtering, radiometric calibration, image stacking, and supervised classification. There are extensive and valuable research works that demonstrated the productive use of modern Geographic Information Systems (GIS), Remote Sensing technologies, and machine learning models to examine LULC dynamics.[4][5][6] Significant studies suggest application of Google Earth Engine capabilities for machine learning based LULC analysis.[7] While several studies have mapped parts of Telangana, research focused on automated LULC classification for regional

level analysis is of great importance for localized mapping, zoning and change detection using modern geospatial technologies.[8]

Methodology: The methods are chosen to be suitable for optimized performance on Sentinel datasets and efficient classification workflows.

- Satellite Data Processing using Sentinel-2 data.
- Supervised Classification (Random Forest, Maximum Likelihood, SVM or integrated ML classifiers).
- GIS-based Zoning using pre-processed outputs, further refined in QGIS/ArcGIS.

Data Collection & Sampling:

Data Sources:

Sentinel-2 Level-2A multispectral optical imagery of the study area are obtained from the Copernicus Data Space Ecosystem [9], for the required period, using the AOI shapefile (Area of Interest) and filter options. The study area is to be defined using the boundary shapefile for the region of interest, and scenes with less cloud cover are selected to ensure data quality.

Pre-processing Steps:

- Radiometric and geometric correction
- Atmospheric correction using Sen2Cor
- Band stacking and resampling
- ROI (Region of Interest) extraction for area of study

Training Samples:

- Digitized training polygons for each land class
- Verification using Google Earth imagery

Sampling Strategy: Stratified sampling is done across all major land cover categories to ensure balanced representation.

Classification Workflow:

1. Feature Extraction
 - Computation of remote sensing indices - NDVI, NDWI, NDBI, EVI, SAVI
 - Texture analysis using GLCM (Gray-Level Co-occurrence Matrix)
2. Training Data Collection
 - Ground truth via Google Earth Pro
 - Sample polygons for each class
3. Classification Techniques
 - Supervised classification using Random Forest and SVM
 - Unsupervised classification using K-Means (if needed)
 - Zonal statistics for mandal-level summaries

Required Tools & Software:

- QGIS / ArcGIS
- Google Earth Pro
- Google Earth Engine
- ESA SNAP Toolbox
- Python
- Computing system with sufficient RAM

Expected Outcomes:

- A high-resolution LULC map for area under research.
- Zoning maps with identification of major categories like agricultural, urban, forest, water, and barren zones.

- Change detection maps using multi-temporal data.
- Integration ready datasets for use in QGIS/ArcGIS planning systems. (GeoTIFF file, Shapefile)
- Comprehensive report constituting technical documentation, visual maps, metadata and analytical insights.
- Improved understanding of land-use dynamics for policymakers.

Feasibility of the project:

- The required tools and technologies are free, open-source, and supports automated workflows.
- Sentinel data is readily available at no cost.
- Computing requirements are moderate.
- Classification, zoning, & output visualization can be completed within the academic project timeline.

Project Phases:

Data Collection	Download Sentinel-2 data & collect region boundary shapefile.
Preprocessing	Run atmospheric correction (Sen2Cor), cloud masking, SAR preprocessing (orbit correction, calibration, speckle filtering, terrain correction), sub-setting.
Feature Engineering	Generate NDVI, NDWI, NDBI, EVI. Extract SAR features. Prepare band stacks.
Classification	Prepare training samples, run Random Forest/SVM supervised classification & refine results.
Accuracy Assessment	Confusion matrix, Kappa coefficient, validation using Google Earth Pro/field samples.
Zoning & Map Generation	Create final LULC maps, zoning layers, mandal-wise statistics. Export GeoTIFF and shapefiles.
Documentation	Consolidate the results, prepare final project report & metadata.

Conclusion: This research project develops an accurate and scientifically validated framework for land use and land cover classification, utilizing open-source satellite data, and advanced geospatial technologies. The outcome of this study will provide valuable support for regional planning, sustainable resource management, and data-driven governance.

List of Abbreviations:

Abbreviation Full Form

LULC	Land Use and Land Cover
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NDBI	Normalized Difference Built-up Index
EVI	Enhanced Vegetation Index
SAVI	Soil Adjusted Vegetation Index
SNAP	Sentinel Application Platform
GIS	Geographic Information System
SAR	Synthetic Aperture Radar

References:

1. Jancevičius, J. (2025). Enhancing land use classification with hybrid machine learning and satellite imagery. *New Trends in Computer Sciences*, 3(1), 1–17. <https://doi.org/10.3846/ntcs.2025.23701>
2. Mahendra, H. N., Pushpalatha, V., Rekha, V., Sharmila, N., Kumar, D. M., Pavithra, G. S., Basavaraj, N. M., and Mallikarjunaswamy, S., 2025. Land Use/Land Cover (LULC) change classification for change detection analysis of remotely sensed data using machine learning-based random forest classifier. *Nature Environment and Pollution Technology*, 24(2), p. B4238. <https://doi.org/10.46488/NEPT.2025.v24i02.B4238>.
3. Misheck Lesa Chundu, Kawawa Banda, Chisanga Lyoba, Greyfold Tembo, Henry M. Sichingabula, Imasiku A. Nyambe, Modeling land use/land cover changes using quad hybrid machine learning model in Bangweulu wetland and surrounding areas, Zambia, *Environmental Challenges*, Volume 14, 2024, 100866, ISSN 2667-0100. <https://doi.org/10.1016/j.envc.2024.100866>.
4. Remote Sensing and GIS for LULC Change Detection: A Review of Techniques and Findings Reema Lahane, Dr. Ramesh R. Manza, Sujata Ambhore, Shital Katkade. <https://doi.org/10.22214/ijraset.2025.72941>
5. Saakshi, Bhardwaj, D.R. & Bhatia, A.K. Land use and land cover analysis in agroforestry and other land use categories in Shimla district of Himachal Pradesh: a GIS-based analysis in North Western Himalayas. *Discov. For.* 1, 33 (2025). <https://doi.org/10.1007/s44415-025-00038-9>
6. Multi-Temporal Image Processing for LULC Classification and Change Detection Saket Raj, Kishan Singh Rawat, Vinod Kumar Tripathi. <https://doi.org/10.60151/envec/RODF5502>
7. Loukika, K. N., Keesara, V. R., & Sridhar, V. (2021). Analysis of Land Use and Land Cover Using Machine Learning Algorithms on Google Earth Engine for Munneru River Basin, India. *Sustainability*, 13(24), 13758. <https://doi.org/10.3390/su132413758>
8. Andalu, A., Naik, M.G. & Budde, S. Spatio-temporal analysis of land use and land cover (LULC) changes using an object-based image analysis (OBIA) approach: a case study of Telangana state agglomeration, India. *GeoJournal* 90, 135 (2025). <https://doi.org/10.1007/s10708-025-11394-6>
9. <https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-data/sentinel-2>

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Student's Signature

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